Taylor series

The Taylor series expansion of $f(x)$ about a:

$$f(a)+f'(a)(x-a)+\frac{f^{(2)}(a)}{2!}(x-a)^2+\frac{f^{(3)}(a)}{3!}(x-a)^3+\ldots$$

Solving non-linear equation:

Bisection method	$x_m = \frac{x_\ell + x_u}{2}$
False-position method	$x_m = x_u - \frac{f(x_u)(x_L - x_U)}{f(x_L) - f(x_U)}$
Fixed-point iteration	$x_k = g(x_{k-1})$
Newton-Raphson method	$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$
Secant method	$x_{i+1} = x_i - \frac{f(x_i)(x_i - x_{i-1})}{f(x_i) - f(x_{i-1})}$
Modified Secant method	$x_i - \frac{\delta\, x_i f(x_i)}{f(x_i + \delta\, x_i) - f(x_i)}$

Linear regression

$a_1 = \frac{n\sum x_i y_i - \sum x_i \sum y_i}{n\sum x_i^2 - (\sum x_i)^2}$ $a_0 = \frac{\sum y_i}{n} - a_1 \frac{\sum x_i}{n}$	$S_t = \sum (y_i - \overline{y_{model}})^2$
$r = \sqrt{\frac{S_t - S_r}{S_t}}$	$S_r = \sum_{i=1}^{n} (y_i - y_{model})^2$